



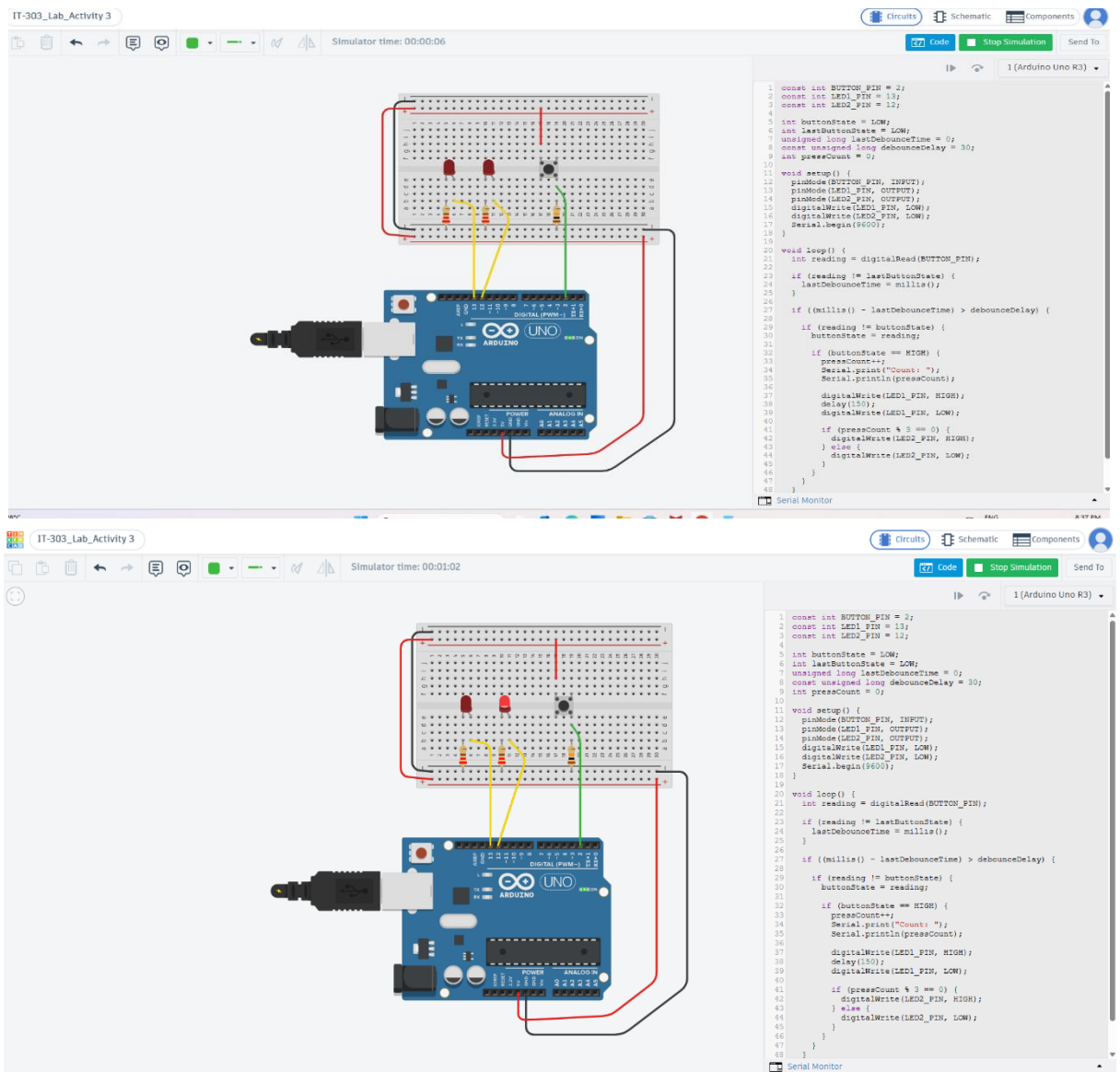
Activity 4: Two architectures, comparison table, justified choice

I. Requirement Statement.

The LED will turn ON when pressing or hold the button, and turn off every release. The system records it each change.

II. Screenshots of both systems working.

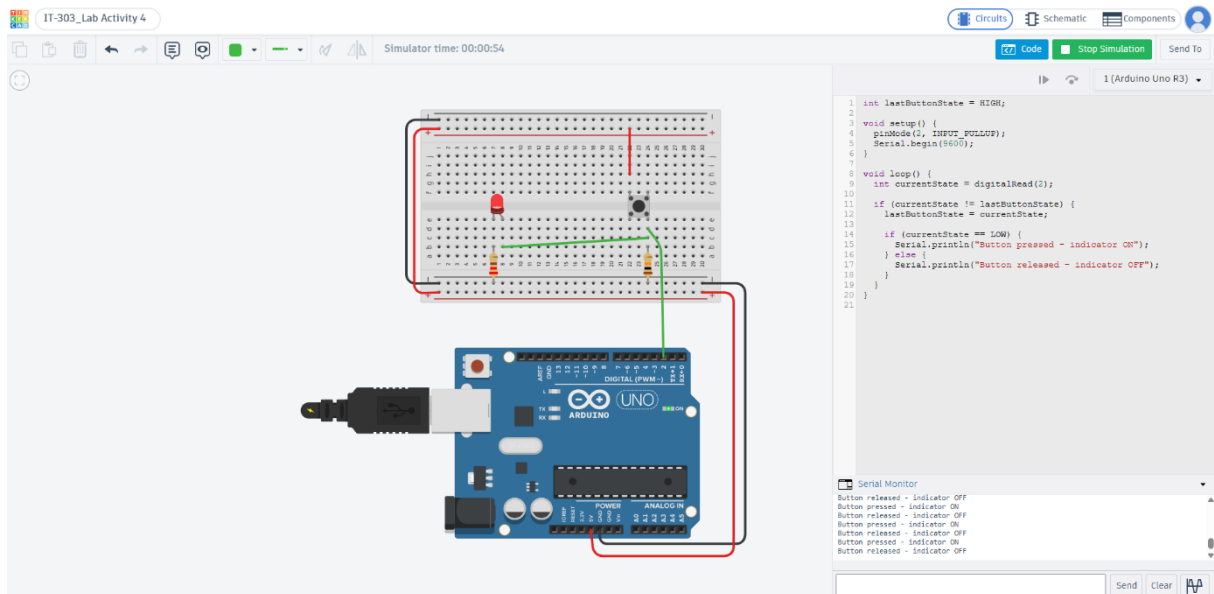
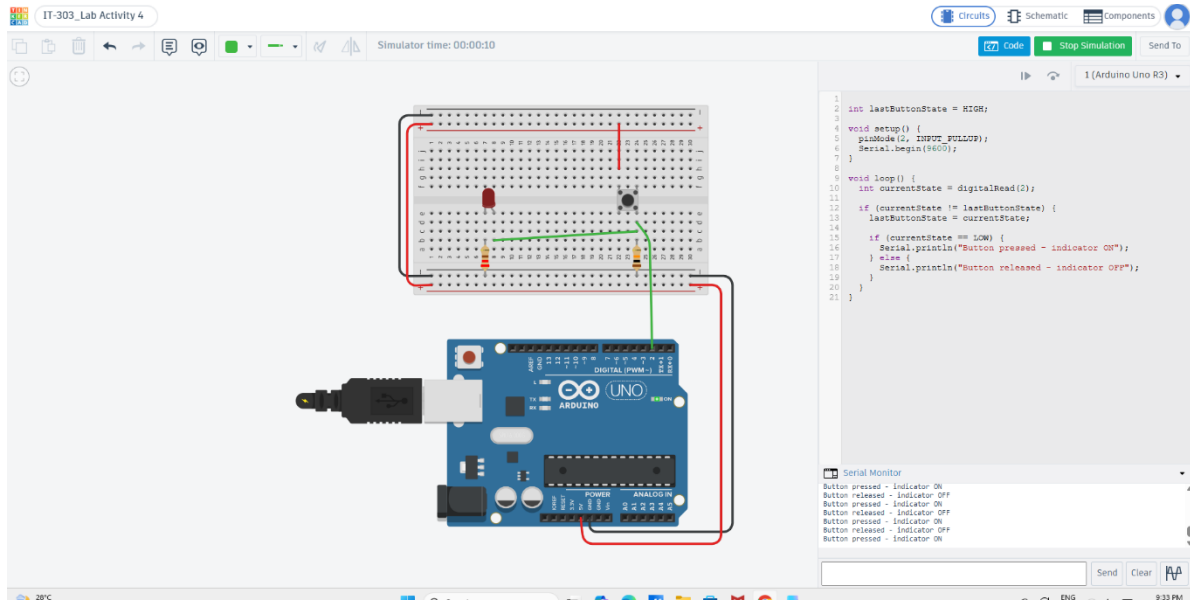
SYSTEM 1 – Activity 3:



ISO 9001:2015



SYSTEM 2 – Activity 4:



ISO 9001:2015



III. The completed comparison table, all four criteria.

Criteria	System 1 – Activity 3 With Arduino controlled	System 2 – Activity 4 Direct to button controlled
Parts Count	Arduino, 1 pushbutton, 2 LEDs, 3 resistors, and connecting wires	Arduino, 1 pushbutton, 1 LED, 1 resistors, and connecting wires
Junctions & Connections	Button to Arduino input, LEDs to Arduino output pins	Button to LED directly; one wire from button only to Arduino input pin
Cost of Change: Make indicator blink	Very easy - update the code only, add delay() and toggle logic. No hardware changes needed.	Difficult - cannot be done through code alone. Requires rewiring and adding components (transistor/relay) since Arduino does not control the LED.
Failure Behaviour: Controller stopped	Entire system fails - LED stays off; button has no effect at all.	LED still works through button turns LED on/off normally and only recording to Serial Monitor stops.





INTERFACE NOTES System 1 – Activity 3

Junction / Interface	Signal / Value	Timing / Behavior	Notes & Evidence
Button to Arduino Pin 2	Digital: HIGH = Pressed, LOW = Released	Changes instantly when button pressed/released; debounce delay: 30ms	Serial Monitor confirms: PRESSED - Count: number / RELEASED
Arduino Pin 13 to LED1	Digital: HIGH = ON, LOW = OFF	ON for 150ms every button press	Blinks briefly on each press - confirmed visually
Arduino Pin 12 to LED2	Digital: HIGH = ON, LOW = OFF	Changes state only on every 3rd press	Remains ON/OFF until next multiple-of-3 press
Arduino to Serial Monitor	Text: PRESSED — Count: N / RELEASED	Updates only on state change — not looping continuously	9600 baud rate, confirmed readable in Serial Monitor





INTERFACE NOTES System 2 – Activity 4

Junction / Interface	Signal / Value	Timing / Behaviour	Notes & Evidence
Button to LED (direct connection)	Power flows directly when pressed — no code involved	Instant ON when pressed; instant OFF when released — zero delay	LED works even when Arduino disconnected
Button to Arduino Pin 2	Digital: HIGH = Pressed, LOW = Released	Reads state change; debounce not required — reporting only	Serial Monitor: PRESSED / RELEASED — reports only, does not control
Arduino to Serial Monitor	Text: PRESSED / RELEASED	Updates on state change — observer/reporter only	9600 baud rate; LED behavior does not depend on this

IV. Your two-sentence justified choice.

If I had to choose between the two systems, I would choose **System 1** because the requirement states that the system must record every change, and System 1 guarantees that every action is both triggered and logged by the controller. Then **System 2** is not a choice because changes can happen without being recorded whenever the controller is not running.

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